

based cloud integrations with the official URL.

InterPlanetary File System (IPFS): Blockchain based Hypermedia Protocol with peer-to-peer communication

There are separate, highly secure file systems and protocols that work with the blockchain environment. An important one for the decentralised/distributed Web/distributed cloud is IPFS or InterPlanetary File System (<https://ipfs.io/>), which is a dedicated protocol for applications.

In traditional Web based environments, the Web application or service is accessed using Hyper Text Transfer Protocol (HTTP), as in <http://www.rediff.com>, <http://www.google.com>, <http://www.du.ac.in>, etc.

In the case of blockchain based file systems, the protocol of IPFS is used with a dynamic hash key with secured encryption, as follows:

- `ipfs://QmUDT1jRPuaiZYky1kfvoMwWdFLqrJiuNDRvMaYc8ohW2`
- `ipfs://QmUDT1jRPuaiZYky1kfvoMwWdFLqrJiuNDRvMaYc8ohW2, etc.`

The use of IPFS makes the entire communication secure and anonymous. The senders and receivers can communicate and share the documents in secured channels of the blockchain.

Table 1

Decentralised platforms	Official URL
IPFS	ipfs.io
Sia	sia.tech
Storj	storj.io
Iagon	iagon.com
Cubbit	cubbit.io
IExec	iex.ec
PPIO	pp.io
Opacity	opacity.io
BTFS	bittorrent.com/token/bittorrent-file-system
NeoFS	fs.neo.org
Dfinity	dfinity.org
Solana	solana.com
Blockstack	blockstack.org
SafeNetwork	safenetwork.tech

IPFS can be installed and implemented with the following:

- IPFS desktop
- Command-line install
- IPFS Companion for browsers
- IPFS Cluster for Big Data and server deployments

The development versions of IPFS are available in variants of the Go language and JavaScript. These developmental versions are used by developers to create and work on private blockchain environments

for corporate and Big Data based applications. Many blockchain based deployments use Go language because of its high performance — fast compilation speed, dynamic error checking, scalability, simplicity, garbage collection, etc.

The desktop version of IPFS is available for multiple operating systems and can be deployed for blockchain based implementations. IPFS can be installed on the Windows operating system by downloading it from <https://github.com/ipfs-shipyards/ipfs-desktop>.

The following commands can be used to install IPFS using various package managers:

Chocolatey: `choco install ipfs-desktop`
 Homebrew: `brew cask install ipfs`
 Snap: `snap install ipfs-desktop`

Cloud integration for blockchain scenarios

The cloud integration of IPFS can be implemented using the Ethereum blockchain platform. This will ensure a higher degree of security in overall communication networks. Once IPFS is integrated, the dashboard is displayed and is used to share the files on the blockchain network, as shown in Figure 5.

Using this approach, the entire communication in an IPFS based blockchain environment can be fully

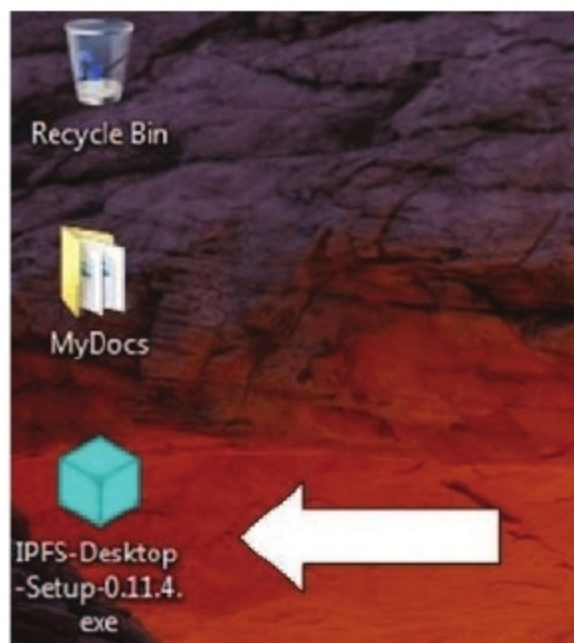


Figure 2: Official portal of IPFS



Figure 3: View of downloaded IPFS installer in Windows